

Assumptions Of ANOVA And Follow Up Procedures

PSYC5141: Experimental Methods for Psychology

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Week 2, Lecture 2

Learning Objectives

- Assumptions of ANOVA
- Rogue data and outliers
- Follow up procedures
- Planned comparisons
- Post-hoc comparisons

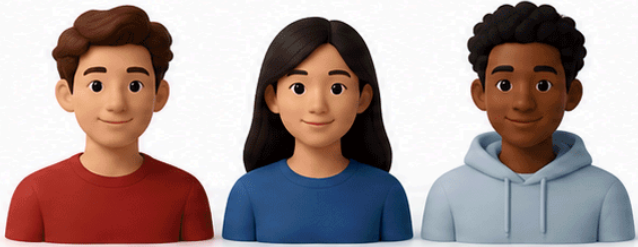
Assumptions of ANOVA

Why Does ANOVA Have Assumptions?

- ANOVA allows us to use our sample data to draw conclusions about populations
- To do this, the F -test makes certain assumptions about our data
- If these assumptions are seriously violated, our statistical conclusions may be less trustworthy
- Fortunately, ANOVA is reasonably robust to modest violations of some assumptions

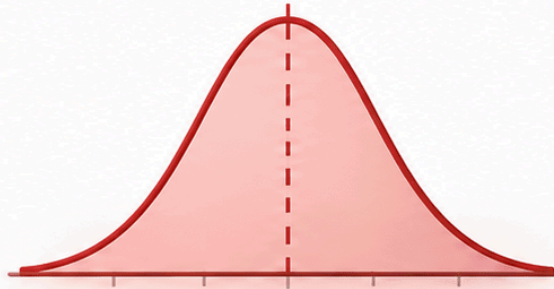
Before interpreting an ANOVA, we should consider whether its assumptions are reasonably satisfied

Assumptions Of ANOVA



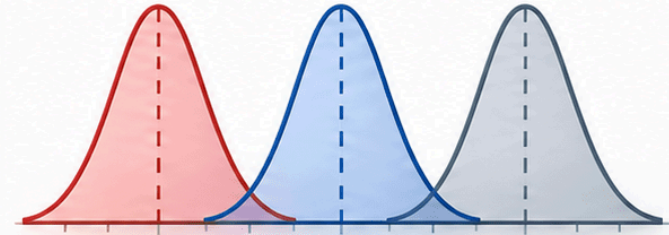
1. Independence of observations

Observations should be independent of one another.



2. Normality

Scores should be approximately normally distributed within each group.

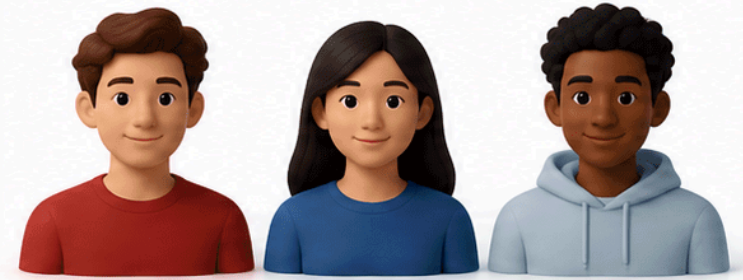


3. Homogeneity of variance

Variability should be reasonably similar across groups.

1. Independence of Observations

- Each observation should be **independent of the others**
- A participant's score should not influence or depend upon another participant's score
- In a between-participants design, each participant should contribute data to only one treatment condition
- Independence is primarily ensured through good experimental design

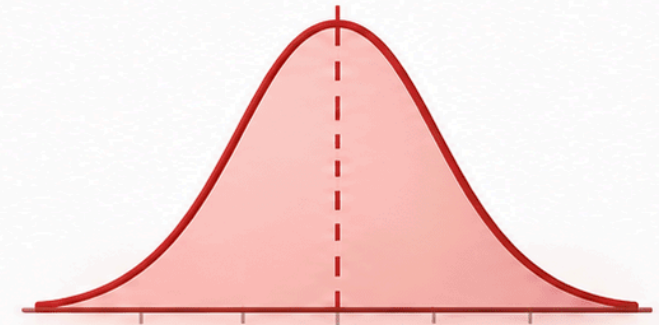


1. Independence of observations

Observations should be independent of one another.

2. Normality

- Scores should be approximately normally distributed within each treatment group
- The assumption concerns the distributions **within groups**, not simply the distribution of all scores combined
- Small departures from normality are generally less concerning than substantial skew or extreme observations

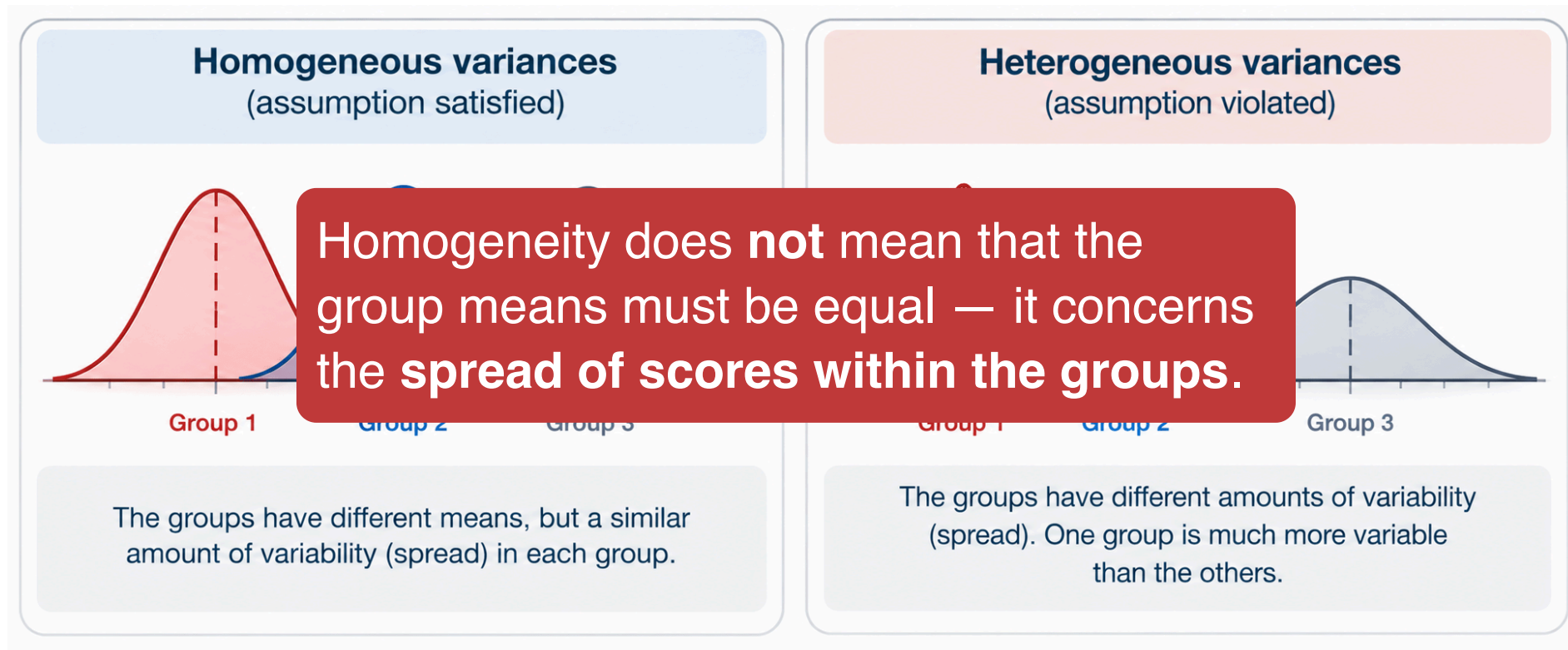


2. Normality

Scores should be approximately normally distributed within each group.

3. Homogeneity of Variance

- The amount of variability should be reasonably similar across treatment groups
- In other words, one group should not be much more variable than another
- This is the assumption of **homogeneity of variance**



Assumptions and Rogue Data

- Normality and homogeneity of variance can both be affected by unusual or extreme observations
- Before conducting an ANOVA, we therefore need to examine our data for potential problems
- This brings us to the issue of **rogue data** and **outliers**

Rogue Data and Outliers

When Data Causes Problems

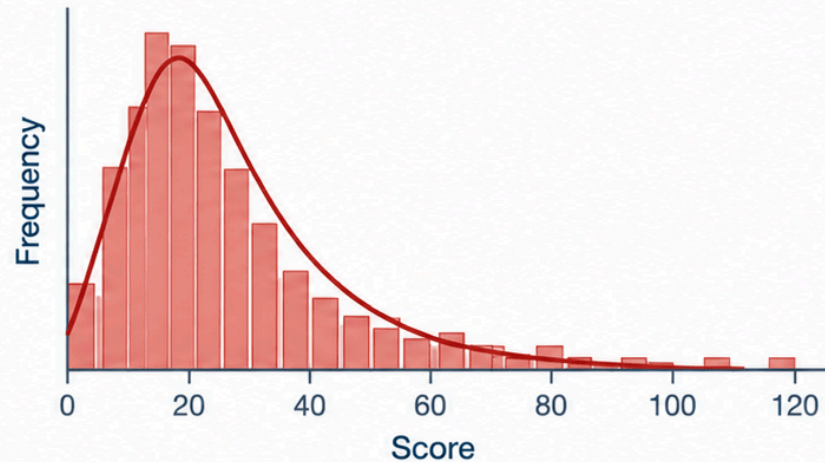
- Real datasets do not always conform neatly to the assumptions of ANOVA
- Problems can include skewed distributions, unequal variances, and extreme observations
- Before analysing the data, we should consider why the problem has occurred
- Possible responses include transforming the data, dealing appropriately with outliers, or—in more serious cases—using a different analysis

Transforming Data

- Transformations change the scale on which the data are expressed
- They can sometimes reduce skew, lessen the influence of extreme scores, or improve homogeneity of variance
- Common examples include logarithmic, square-root, and reciprocal transformations
- The transformed scores—not the original scores—are then used in the analysis

Transforming Data

Original data
(positively skewed)



The data are positively skewed, with a long tail of high scores.

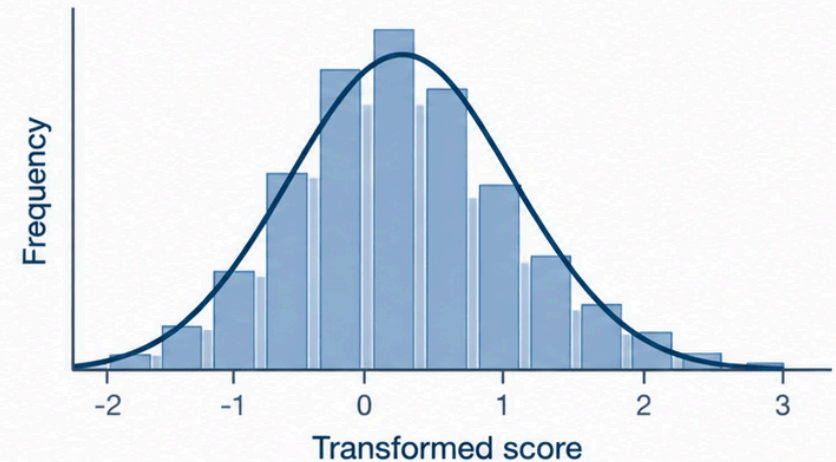
Transformation

Apply the same mathematical function to every score

Examples

- Logarithmic transformation
- Square-root transformation
- Reciprocal transformation

Transformed data
(more symmetric)



After transformation, the distribution is more nearly symmetric and the influence of extreme scores is reduced.

Limitations of Transformations

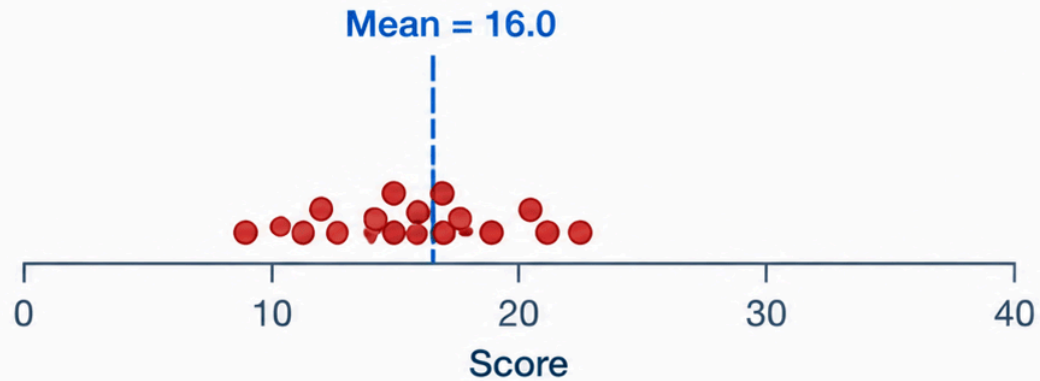
- A transformation does not guarantee that assumption violations will disappear
- It may not help when different groups are skewed in different directions
- Transformations make results less intuitive to interpret because the analysis is conducted on a different scale
- Because ANOVA is relatively robust, transformation is not automatically necessary for modest departures from its assumptions

Outliers

- An outlier is an observation that is unusually distant from the other observations
- Outliers may arise from measurement/recording errors, lapses in attention, or genuine individual differences
- Extreme scores can disproportionately affect the mean and variance
- An extreme observation should not automatically be deleted

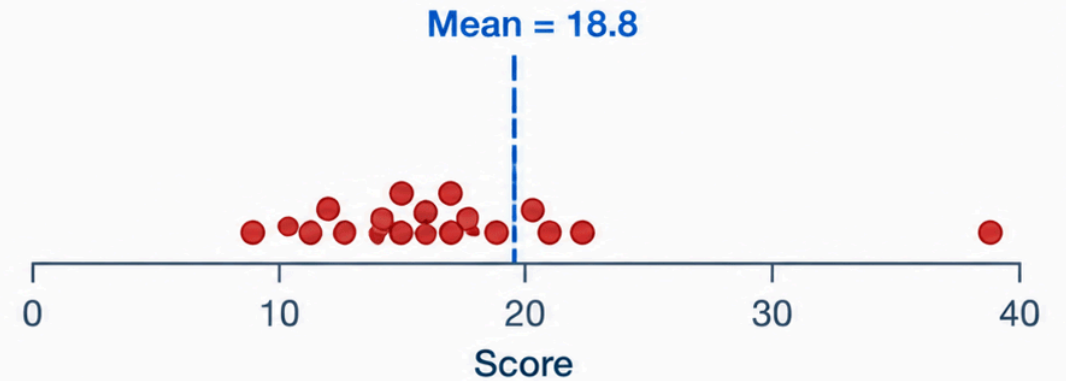
Outliers

Without an outlier



Mean = 16.0
Standard deviation = 4.3
 $n = 15$

With an outlier



Mean = 18.8
Standard deviation = 9.7
 $n = 16$

Dealing With Outliers

- Check the observation: is it a data-entry, coding, or measurement error?
- Consider whether there is a substantive explanation for the extreme value
- Use consistent, defensible criteria when identifying potential outliers
- Any exclusions should be justified and reported transparently

In the practical: we will look at how to identify and deal with potential outliers in R

What If The Data Are Problematic?

- First, consider why the data are problematic
- Check for data errors and extreme observations, and decide how they should be handled
- In some circumstances, a data transformation may improve the properties of the data
- Alternatively, we can use a **non-parametric test**, which makes fewer assumptions about the distribution of the data
- For a single-factor between-participants design, one alternative is the **Kruskal–Wallis test**

Coming in Week 4: We will introduce non-parametric tests and learn when and how to use them

Follow Up Procedures

What Does a Significant ANOVA Tell Us?

- Our ANOVA was significant:

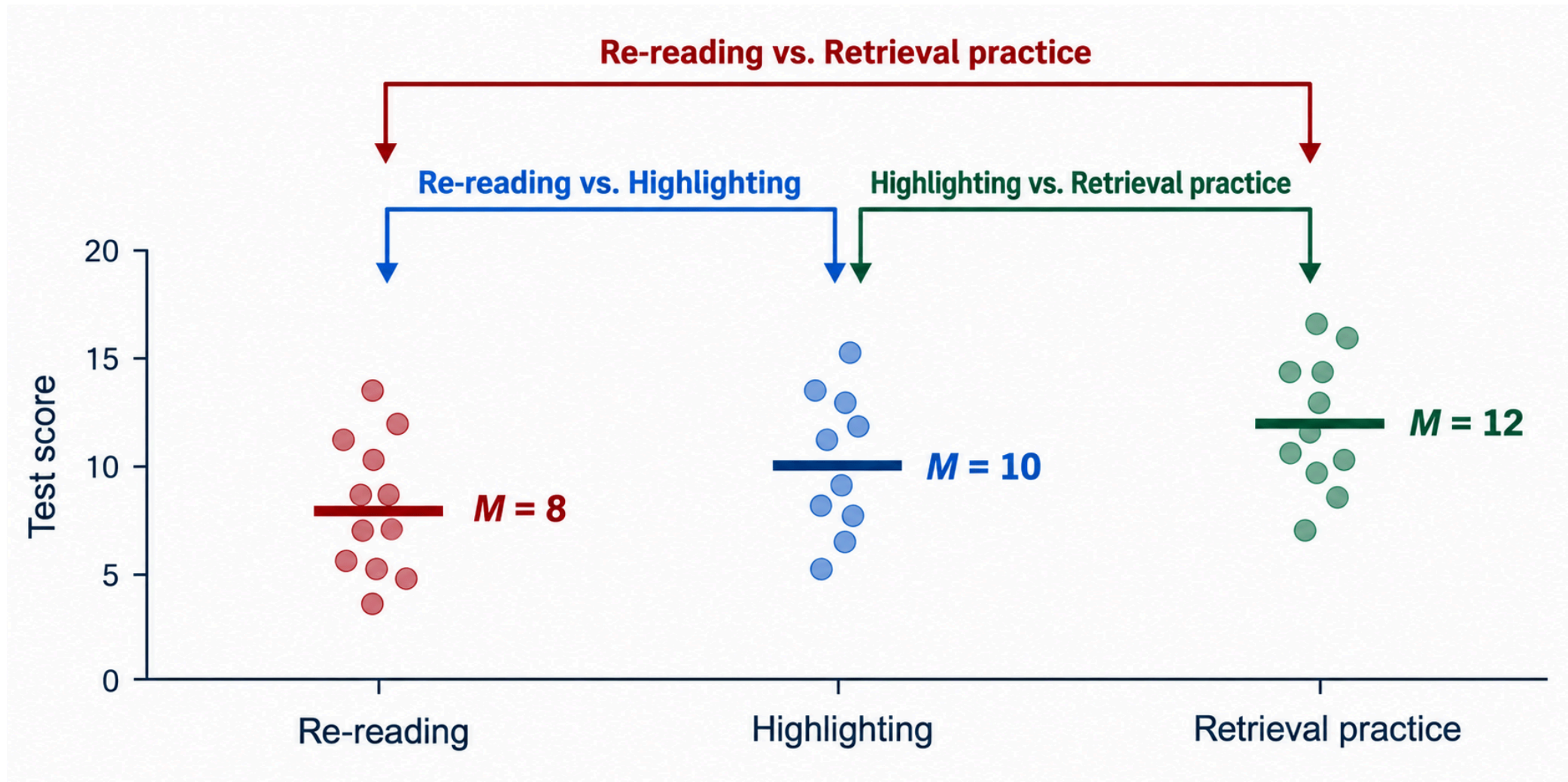
$$F(2, 12) = 5.00, p < .05$$

- We can reject the null hypothesis that all three population means are equal
- At least one population mean differs
- But the omnibus ANOVA doesn't tell us which means differ

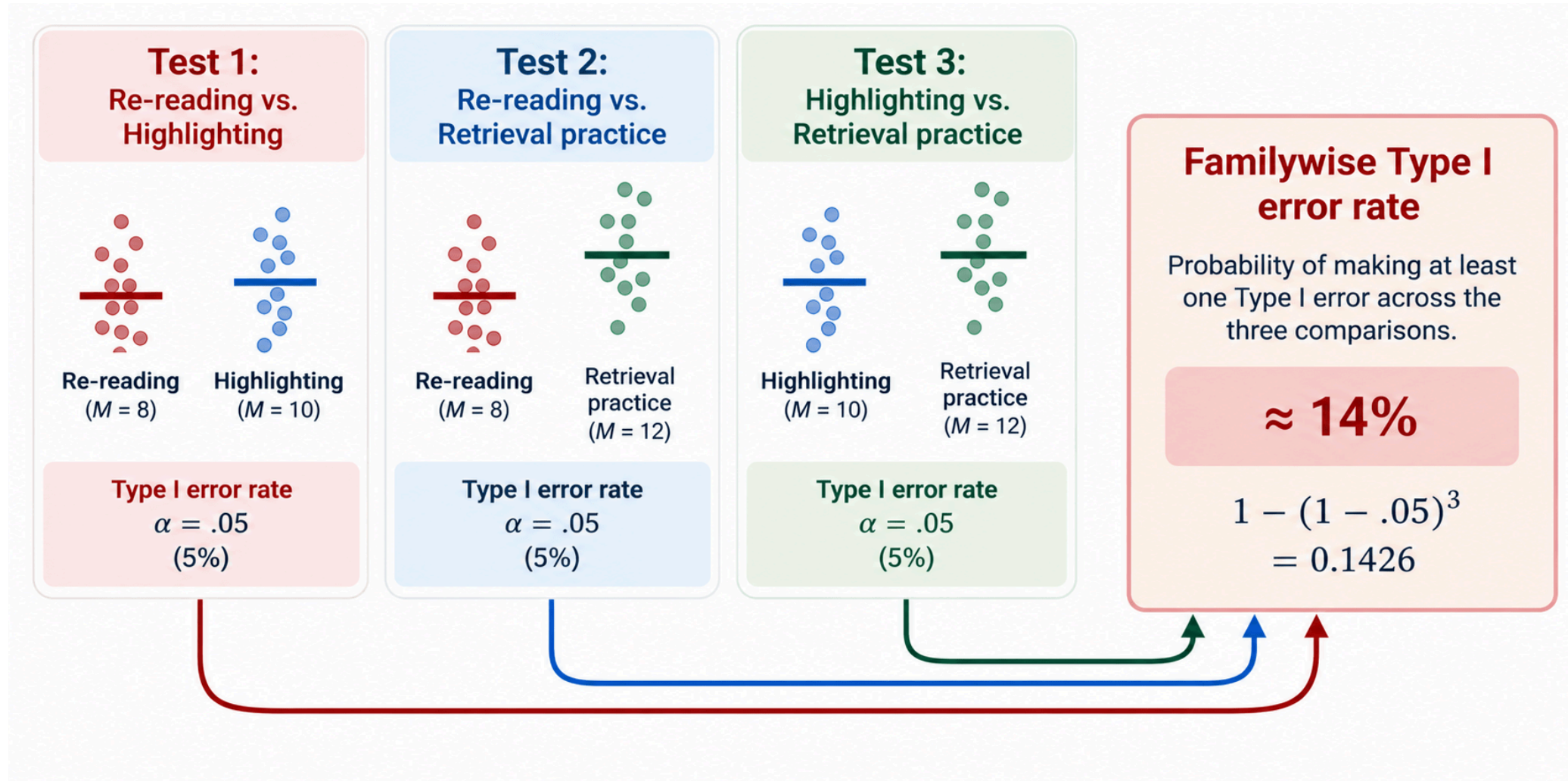
Where does the difference lie?

Re-reading	Highlighting	Retrieval practice
8	10	12

Pairwise Comparisons



Why Not Just Run Multiple *t*-Tests?



Two Approaches to Follow Up Comparisons

1. Planned comparisons

- Comparisons specified **before examining the results**
- Used to test particular hypotheses

2. Post-hoc comparisons

- Comparisons decided upon **after examining the results**
- Used to explore where differences lie

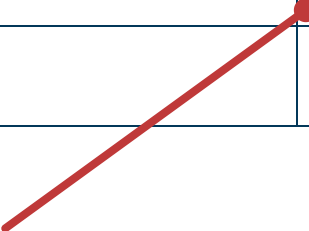
Planned Comparisons

Planned Comparisons

- Planned comparisons test specific hypotheses identified in advance
- We do not need to compare every possible pair of means
- Suppose we predicted that retrieval practice would outperform both other study strategies
- **Hypothesis 1:** Retrieval practice > Re-reading
- **Hypothesis 2:** Retrieval practice > Highlighting

Planned Comparisons

Source of variance	Sum of squares	Degrees of freedom	Variance (Mean square)	<i>F</i>
Between groups	40	2	20	5.00
Within groups	48	12	4	
Total	88	14		


$$MS_{\text{error}} = MS_{\text{within}} = 4$$

In the previous lecture, we calculated the within-group variance as our estimate of experimental error. We can now use that same estimate of experimental error when testing our planned comparisons

Testing a Planned Comparison

The two means being compared

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{MS_{\text{error}} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

The within-group mean square from the ANOVA

The number of observations contributing to those means

The diagram shows the t-test formula for a planned comparison. Red arrows point from descriptive text to parts of the formula: one arrow points from 'The two means being compared' to the numerator $\bar{X}_1 - \bar{X}_2$; another arrow points from 'The within-group mean square from the ANOVA' to MS_{error} ; and two arrows point from 'The number of observations contributing to those means' to n_1 and n_2 in the denominator.

Planned Comparison 1: Retrieval Practice vs. Re-Reading

$$\bar{X}_1 = 12 \quad \bar{X}_2 = 8$$

$$n_1 = n_2 = 5 \quad MS_{\text{error}} = 4$$

and substitute:

$$t = \frac{12 - 8}{\sqrt{4 \left(\frac{1}{5} + \frac{1}{5} \right)}}$$

$$t = \frac{4}{\sqrt{1.6}} = 3.16$$

Planned Comparison 1: Retrieval Practice vs. Re-Reading

We use the error degrees of freedom from our ANOVA

$$|t| = 3.16 > 2.18$$

therefore **reject**(H_0)

Retrieval practice produced significantly higher scores than re-reading

Critical Values of t

Values are for a **two-tailed** test with significance level α .

Degrees of freedom (df)	$\alpha = .10$	$\alpha = .05$	$\alpha = .02$	$\alpha = .01$
1	6.314	12.706	31.821	63.657
2	2.920	4.303	6.965	9.925
3	2.353	3.182	4.541	5.841
4	2.132	2.776	3.747	4.604
5	2.015	2.571	3.365	4.032
6	1.943	2.447	3.143	3.707
7	1.895	2.365	2.998	3.499
8	1.860	2.306	2.896	3.355
9	1.833	2.262	2.821	3.250
10	1.812	2.228	2.764	3.169
11	1.796	2.201	2.718	3.106
12	1.782	2.179	2.681	3.055
13	1.771	2.160	2.650	3.012
14	1.761	2.145	2.624	2.977
15	1.753	2.131	2.602	2.947
20	1.725	2.086	2.528	2.845
30	1.697	2.042	2.457	2.750
60	1.671	2.000	2.390	2.660
∞	1.645	1.960	2.326	2.576

Planned Comparison 2: Retrieval Practice vs. Highlighting

$$\bar{X}_1 = 12 \quad \bar{X}_2 = 10$$

$$t = \frac{12 - 10}{\sqrt{4 \left(\frac{1}{5} + \frac{1}{5} \right)}} = \frac{4}{\sqrt{1.6}} = 1.58$$

Because $df = 12$, we already know:

$$|t| = 1.58 < 2.18$$

$\therefore$ do not reject H_0

Retrieval practice did not
produce significantly higher
scores than highlighting

Summary

Retrieval practice vs. Re-reading: significant ✓

Retrieval practice vs. Highlighting: not significant ✗

A **significant** omnibus ANOVA **does not** mean that every pair of means differs significantly.

Do Planned Comparisons Need a Correction?

- Multiple comparisons increase the **familywise Type I error rate**
- Keep the number of planned comparisons **as low as possible**
- A useful rule of thumb is

$$\text{Maximum planned comparisons} = a - 1$$

- With three levels:

$$3 - 1 = 2 \text{ planned comparisons}$$

With a small number of genuinely planned comparisons, we can test each at the usual $\alpha = .05$. With larger numbers, we may need to adjust our significance level

When Might We Use Bonferroni?

- With larger numbers of comparisons, familywise Type I error becomes increasingly problematic
- One simple way to control this is the **Bonferroni correction**
- Divide the desired familywise significance level by the number of comparisons:

$$\alpha_{\text{adjusted}} = \frac{\alpha}{\text{number of comparisons}}$$

- Example:

$$\frac{.05}{5} = .01$$

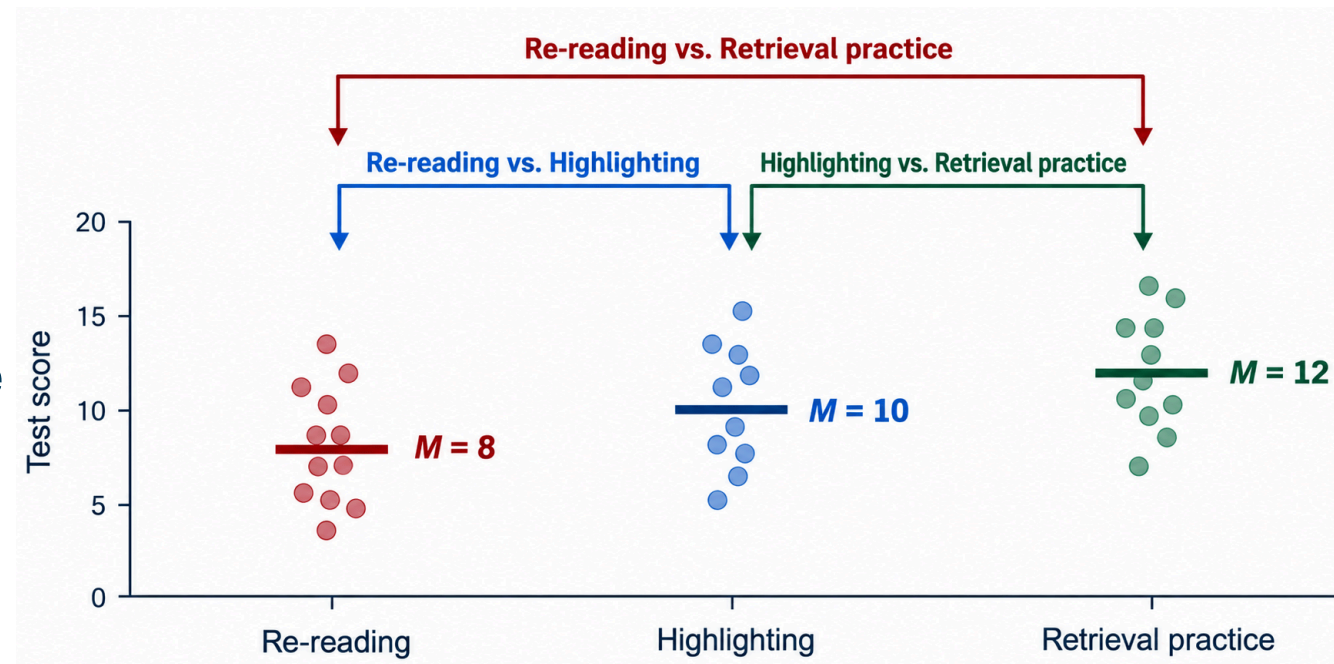
With five comparisons, each test must therefore reach $p < .01$.

Post-Hoc Comparisons

When Do We Use Post-Hoc Tests?

- Sometimes we have no specific predictions about which means will differ
- Following a significant omnibus ANOVA, we may want to determine where the differences lie
- We therefore examine **all possible pairwise comparisons**
- But making multiple comparisons increases the **familywise Type I error rate**
- **Post-hoc tests** allow us to make these comparisons while controlling that error rate

Post-hoc tests examine all possible pairwise comparisons



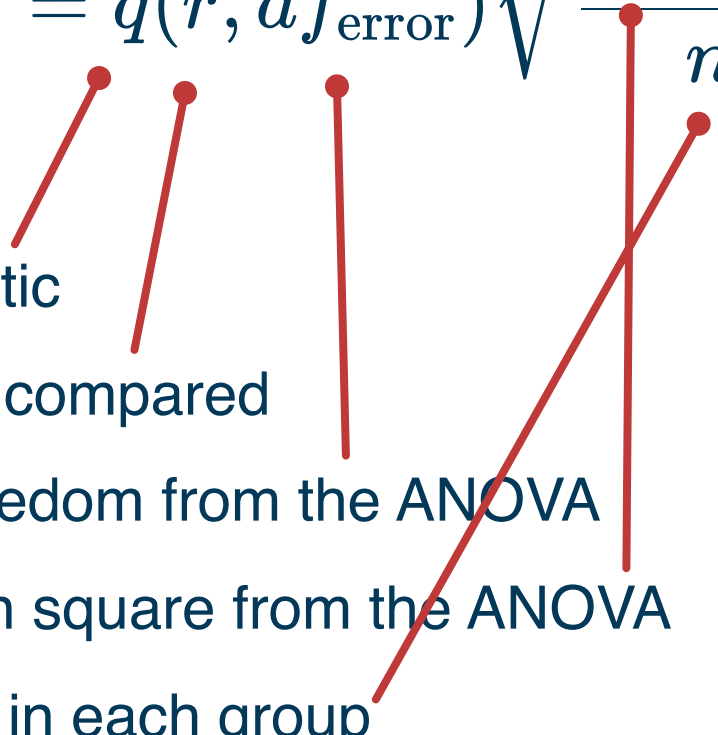
Tukey-Kramer HSD

- One commonly used post-hoc procedure is the **Tukey–Kramer HSD**
- HSD stands for **Honestly Significant Difference**
- It determines the **minimum difference between two means** required for statistical significance
- It controls the **familywise Type I error rate** across the set of pairwise comparisons

Observed difference between means $> W \Rightarrow$ significant

where W is the **critical difference**

Calculating The Critical Difference

$$W = q(r, df_{\text{error}}) \sqrt{\frac{MS_{\text{error}}}{n}}$$


- q = Studentized range statistic
- r = number of means being compared
- df_{error} = error degrees of freedom from the ANOVA
- MS_{error} = within-group mean square from the ANOVA
- n = number of observations in each group

Just like our planned comparisons, Tukey uses the estimate of experimental error from our ANOVA

Applying Tukey-Kramer HSD

- From our ANOVA:

$$r = 3$$

$$df_{\text{error}} = 12$$

$$MS_{\text{error}} = 4$$

$$n = 5$$

- We then need the appropriate **Studentized range statistic** q .

Finding the Critical Value of q

Number of levels of factor

Error degrees of freedom
from our ANOVA

The relevant q value is
approximately **3.77**.

Then substitute:

$$W = 3.77 \sqrt{\frac{4}{5}}$$
$$W = 3.77(0.894)$$

$W \approx 3.37$

Two means must differ by more than approx.
3.37 points to be significantly different

Studentized Range (q) Table

Critical values of the studentized range statistic (q) for $\alpha = .05$

Error degrees of freedom (df_{error})	Number of means (r)								
	2	3	4	5	6	7	8	9	10
5	3.64	4.60	5.22	5.67	6.03	6.33	6.58	6.80	6.99
6	3.46	4.34	4.90	5.31	5.63	5.90	6.13	6.33	6.51
7	3.34	4.16	4.68	5.06	5.36	5.61	5.83	6.02	6.19
8	3.26	4.04	4.53	4.89	5.17	5.41	5.61	5.79	5.95
9	3.20	3.95	4.41	4.75	5.01	5.24	5.43	5.60	5.75
10	3.15	3.88	4.33	4.65	4.91	5.12	5.30	5.46	5.61
11	3.11	3.82	4.25	4.56	4.80	5.01	5.18	5.34	5.48
12	3.08	3.77	4.18	4.48	4.71	4.91	5.08	5.23	5.37
13	3.06	3.73	4.12	4.41	4.63	4.82	4.99	5.13	5.26
14	3.03	3.70	4.08	4.36	4.58	4.76	4.92	5.06	5.19
15	3.01	3.67	4.04	4.32	4.53	4.71	4.86	5.00	5.12
20	2.97	3.58	3.93	4.18	4.37	4.53	4.67	4.80	4.91
24	2.94	3.52	3.86	4.10	4.28	4.43	4.56	4.68	4.79
30	2.92	3.49	3.81	4.04	4.21	4.36	4.48	4.60	4.70
40	2.89	3.44	3.75	3.98	4.14	4.28	4.40	4.51	4.61
60	2.86	3.40	3.70	3.92	4.08	4.22	4.33	4.44	4.53
∞	2.77	3.31	3.60	3.82	3.98	4.11	4.22	4.33	4.42

Note: Values are for $\alpha = .05$. Use the value corresponding to the number of means (r) and the error degrees of freedom (df_{error}). 39

Which Means Differ?

Comparison	Mean difference	Critical difference	Significant?
Retrieval practice – Re-reading	$12 - 8 = 4$	3.37	Yes ✓
Retrieval practice – Highlighting	$12 - 10 = 2$	3.37	No ✗
Highlighting – Re-reading	$10 - 8 = 2$	3.37	No ✗

Key takeaway: Only retrieval practice and re-reading differ significantly

Follow Up Procedures Summary

Planned comparisons	Post-hoc comparisons
Specified before examining the data	Selected after examining the data
Test specific hypotheses	Explore which means differ
Usually involve selected comparisons	Can examine all pairwise comparisons
Can use <i>t</i> -tests with the ANOVA error term	Can use Tukey–Kramer HSD
Multiple-testing correction may be appropriate	Familywise error control is built into the procedure

Key takeaway: The choice between planned and post-hoc comparisons depends on the research questions and when the comparisons were specified

That's all folks!